

Spectra Sync

Core Feature

Calculate spatial translations between images using phase correlation.

Implementation

func_translation →

fourier shift theorem

↳ dirac delta function
which outputs a spike

↳ find the coordinate

↳ calculate shift

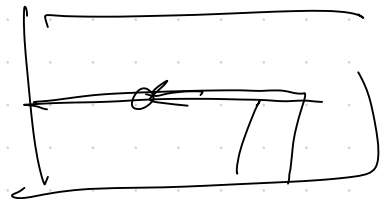
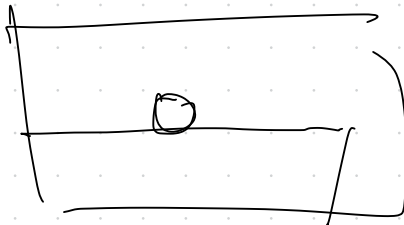
func_rotation and scale alignment → fourier mellin transform

↳ FFT magnitude both image

↳ log polar coordinates

↳ Rotation angle

& scale factor



Applied features

multiframe stacking for noise reduction

methodology:

- use translation for alignment
- median stack pixelwise

moving object removal from video

methodology:

- same as multiframe stacking but different usecase

highlight moving object

- use object removal
- then detect high change
- then detect the outline